E-Waste & Environmental Management

Activity Worksheet

The Hidden Carbon Footprint of Our Gadgets

Name:	Om Thakur	**Roll No.:**	2410100041
Division:	C	**Date:**	5/8/26

Neilima ⑦ → ⑯ Neilima

Part A: Carbon Footprint Calculation

Use the carbon footprint calculator provided by your faculty.

Parameter	Your Response
Annual Carbon Footprint	34 tCO_2e / year
National Average	7 tCO_2e / year
Global Average	19 tCO_2e / year
Number of Earths Required (if provided)	~~4~~ 1.8 Earths

Please Read out

Which category contributed the most?

☑ Transportation ☐ Home Energy ☐ Food ☐ Shopping & Consumption

☐ Electronics / Digital Lifestyle ☐ Other: ____________

Which result surprised you the most?

Food. I was surprised that my daily food consumption and dietary choices contributed more to overall emissions despite transportation being my highest category.

Part B: My Electronic Lifestyle

1. How many smartphones have you owned in the last 10 years?

☐ 1 ☑ 2–3 ☐ 4–5 ☐ More than 5

2. How often do you replace your smartphone?

☐ Every year ☐ Every 2 years ☐ Every 3–4 years ☑ More than 5 years

3. What do you usually do with your old phone?

☐ Keep ☐ Sell ☑ Exchange ☐ Donate

☐ Recycle ☐ Throw away

4. Have you ever repaired an electronic device instead of replacing it?

☑ Yes ☐ No

If yes: I changed my phone's battery and display

5. Have you ever recycled e-waste through an authorized recycler?

☐ Yes ☑ No

Item: ______

Unused Electronic Items at Home	Number
Mobile phones	3
Chargers	2
Earphones	2
Batteries	1
Cables	7
Other	

Part C: Think Like an Environmental Engineer

Habit	Higher Carbon Footprint	More E-Waste
Buying a new phone every 2 years	☐	☑
Repairing a laptop	☐	☑
Keeping old phones unused in drawers	☐	☑
Recycling electronic devices	☑	☐
Using devices for 5–6 years	☑	☐

Which stage of an electronic product's life cycle contributes the MOST carbon emissions?

☐ Raw material extraction ☑ Manufacturing ☐ Transportation

☐ Usage ☐ Disposal

Why is extending the life of an electronic device environmentally beneficial?

Extending a device's lifespan delays the need to manufacture a replacement, directly preventing the high energy consumption, raw material extraction, and carbon emissions associated with new production.

Part D: Reflection

Three habits that increase my personal carbon footprint:

1) Reliance on fossil-fueled vehicles
2) Leaving Laptop on charge the whole day
3) Meat based diet.

Three habits that increase e-waste generation:

1) Hoarding old electronics
2) Throwing away cells & batteries
3) Buying gadgets with similar function or ~~usef~~ ~~useful~~ useless currently

Action to Reduce Carbon Emissions & E-waste	Expected Impact
Carpooling to travel to places.	~~It~~ Reduce Carbon emissions
Using electric vehicles instead of fossil-fueled	Reduce Carbon emissions
~~Using~~ Repairing electronic devices	Reduce e-waste
Using biodegradable products often.	Reduce overall waste in general

Discussion Questions (Optional)

1. Which stage of an electronic product's life cycle has the highest carbon footprint and why?

Manufacturing phase like raw material extraction uses processes like silicon refining, chip fabrication require massive amts of energy, high purity chemicals & machinery running continuously.

2. How does extending the lifespan of smartphones reduce both carbon emissions and e-waste?

Extending smartphone lifespans delays the need for new device manufacturing, which directly cuts down on high production emissions and reduces landfills too.

3. Why is recycling alone not enough for sustainable e-waste management?

Recycling alone is insufficient because the process itself consumes high amount of energy and cannot recover 100% of original materials.

4. Explain how the 5Rs contribute to lowering the environmental impact of electronic products.

5Rs (Refuse, Reduce, Reuse, Repair & Recycle) lower environmental impact by curbing overconsumption & keeping materials in circulation longer.

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5. As an engineering student, what innovative solution would you propose to reduce the carbon footprint associated with electronic devices?

I would propose designing modular, ultra-repairable electronics combined with AI-driven energy optimization software to drastically extend product lifespans & reduce manufacturing demand. Integrating Bio-based PCB helps further.

Ms. Nilima Main
Assistant Professor, Department of First Year Engineering
Vidyalankar Institute of Technology